

Fort Saskatchewan Path₂Zero Project - UPI

Standard Purlin Design

FS-D000-CAL-20005

P.Eng Stamp	Permit to Practice Stamp
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Checking Level: 2B

Doc Rev	Revision Description/ Status	Date Issued	Fluor Originator	Checker	Approval
				Fluor Engineering Checker	Fluor Discipline Lead
0	IFI	1-Nov-2023	G. IP	A. SHARIF	K. TULLU

Risk Notice

Does this Calculation have any assumptions or risk items that need to be followed up with?

☐ Yes ☒ No

If Yes, anticipated date the information will be available: _____

Followed up by: _____ Date: _____ Signature: _____

Revision History

Description of revision, especially to original calculation

Rev No	Originator (Name)	Checker (Name)	Revision Date	Revision Description
0	Gary Ip	Alexander Sharif	1-Nov-2023	IFI - Issued for Information

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1. RISK AND UNCERTAINTY LOG

The purpose of the table outlines the uncertainties, mitigation strategy, remaining risk at the time of design.
Assumptions to be re-validated once the appropriate information is made available.

Seq # (PDR # if avail)	Uncertainty	Mitigation	Remaining Risk	Resolved (Y/N)	Resolution (include approx date)

2. COMPUTER FILE INDEX

Title: Standard Purlin Design

Calculation Number: FS-D000-CAL-20005

The purpose of this form is to keep track of calculation computer files and to assign filenames using a standard format. This will assist in organizing data and in retrieving data at a later date. The file naming scheme shown below should be used.

Include this form with each set of calculations which has computer files. List all data files that were used to complete the calculation. Filenames shall be as follows:

FS-AXXX-ZZZ-DDNNN_R.EXT

Where: FS = Project Code
A = Page Size (A - 22x34, B - 17x23, C - 17x11, D - 8.5x11, F - Other)
XXX = Area / Unit Number
ZZZ = Document Type (CAL - Calculations)
DD = Discipline Code (21 - Civil / 20 - Structural)
NNN = Sequence Number
R = Revision Number
EXT = File extension (eg: xls,xlsx, r3d, docx, pdf, etc.)

FILENAME	DESCRIPTION	DIRECTORY LOCATION
FS-D000-CAL-20005_0.xls	Native Excel Calculation File	\\215 STRUCTURAL\215E ENGINEERING DESIGN DATA\215E 014 R Proj Calculations Sketches\WER03\FS-D000-CAL-20005\
FS-D000-CAL-20005_0.pdf	PDF Output	
FS-D000-CAL-20005_0.r3d	RISA Model	
FS-D000-CAL-20005_0 - W10x22 at 15FT.mcdx	Native MathCad Calculation File	
FS-D000-CAL-20005_0 - W10x22 at 20FT.mcdx		
FS-D000-CAL-20005_0 - W10x33 at 15FT.mcdx		
FS-D000-CAL-20005_0 - W10x33 at 20FT.mcdx		
FS-D000-CAL-20005_0 - W10x49 at 15FT.mcdx		
FS-D000-CAL-20005_0 - W10x49 at 20FT.mcdx		
FS-D000-CAL-20005_0 - W10x68 at 15FT.mcdx		
FS-D000-CAL-20005_0 - W10x68 at 20FT.mcdx		
FS-D000-CAL-20005_0 - W12x26 at 15FT.mcdx		
FS-D000-CAL-20005_0 - W12x40 at 25FT.mcdx		

Client: DOW CHEMICAL CANADA ULC

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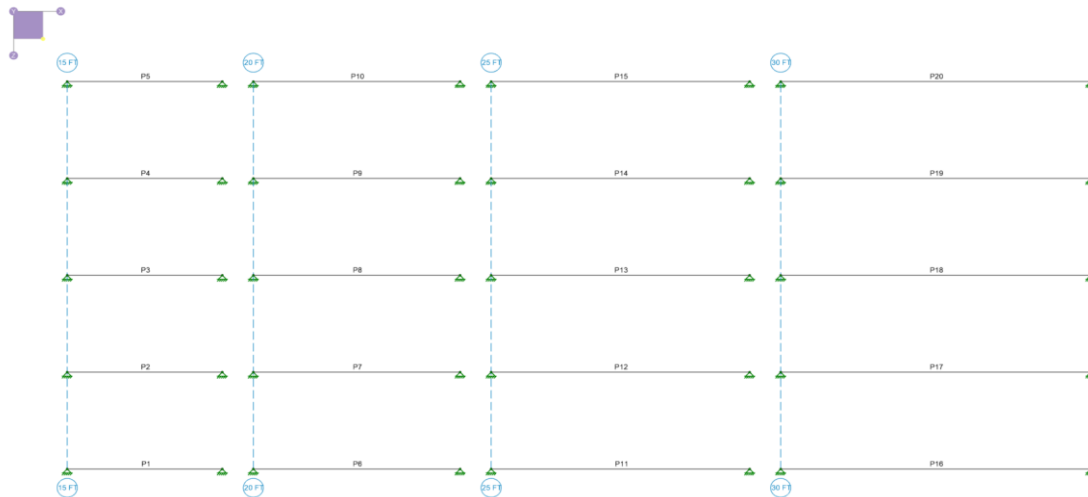
FS-D000-CAL-20005_0 - W12x40 at 30FT.mcdx		
FS-D000-CAL-20005_0 - W12x53 at 25FT.mcdx		
FS-D000-CAL-20005_0 - W12x53 at 30FT.mcdx		
FS-D000-CAL-20005_0 - W12x65 at 25FT.mcdx		
FS-D000-CAL-20005_0 - W12x65 at 30FT.mcdx		
FS-D000-CAL-20005_0 - W14x43 at 25FT.mcdx		
FS-D000-CAL-20005_0 - W14x43 at 30FT.mcdx		

3. DESCRIPTION OF CALCULATION

- This calculation pertains to site-wide standardization of purlin design to assist in selection during detailed design of the various buildings
- Standardized purlin design will consider columns spaced at 15' (4.572 m), 20' (6.096 m), 25' (7.62 m) and 30' (9.144 m), and typical purlin spacing of approx. ~5' (1.524 m), wider spacing must be reconsidered for the additional loadings.
- For the purpose of snow and wind pressure, a maximum bldg. height of 20 m (65.6 FT) is considered. Buildings taller than this must consider additional snow and wind pressure acting on the purlin.

3.1 Description of the Structure

Hot rolled wide-flange beams are considered for purlins. No sag rods are considered. Bolted connections to roof frame beams are designed by others.



- View of the RISA model

4. BASIS OF DESIGN

4.1 Site Location

This calculation pertains to the entire Fort Saskatchewan site located in Alberta.

4.2 Checking Level

The following calculation checking levels are applied to the following calculation. Descriptions for the checking levels are found in the Activity Plan – Civil Structural Engineering.

Check where applicable	Checking Level	Description
☐	1B	Detailed check in accordance with governing work instruction or practice
☒	2B	Selective Check with particular attention to critical and interface points.
☐	3B	Spot check, applied as established by project lead
☐	4B	Self Check only

4.3 Codes and Specifications

4.3.1 Governing Codes

NBC (AE) 2019 – National Building Code - Alberta Edition 2019
NBCC 2015 - Structural Commentaries
CAN/CSA-S16-14 Limit States Design of Steel Structures 11th edition, 2017

4.3.2 Specifications

FS2Z-0001-01	Civil Engineering Design Criteria Global
FS2Z-0001-33	Site Variations to Civil Eng Criteria Alberta Operations
G3Z-1090-00	Wind Design Engineering Criteria for Buildings
G3Z-1090-02	Structural Design Criteria for Buildings - Global

4.3.3 Other Reference Data

Structural Job Data Book

JDB 2.A.02	CSA Calculations
JDB 2.B.04	BLCs-LCs - Buildings
JDB 2.F.06.	Torsion
JDB 2.G.01	Building Structural Design Guideline
JDB 2.H.01	Wind Loading
JDB 2.H.09	Snow Loading

4.4 Material Properties

Anchor Bolts (UNO):	ASTM F1554 Gr36, as specified by the Engineer
Structural Steel (UNO):	Structural Steel W, HP, S, WT CSA G40.21 Grade 350W or ASTM A992 Grade 50 Structural HSS – CSA G40.21 Grade 350W Class C or ASTM A500/A500M Grade C, minimum yield strength 50 ksi (345 MPa)

Channels and Angles – G40.21 Grade 300W minimum or ASTM A572/A572M Grade 50
Steel Plate and flat bar – G40.21 Grade 300W minimum or ASTM A572/A572M Grade 50
Checkered floor plate material – 1/4" (6 mm) thick ASTM A786/A786M minimum yield strength 35 ksi (240 MPa) (UNO)
Grating and Stair Tread – Welded steel grating w/ 1 1/4" x 3/16" (32x4.8) serrated bearing bars at 1 3/16" (30 mm) centers

Structural Bolts (UNO): High Strength bolts – 3/4" Dia ASTM F3125, Grade A325, Type 1(UNO)

Connection plate: Min. fillet weld size shall be 1/4" (6mm)
All gusset plates and connection plate materials shall be min. 3/8" (10mm) thick

4.6 Software

The following outlines the list of computer software used for analysis. Refer to the Computer File Index for files:

Microsoft Excel	365	
Risa-3D	21.0.0.	(aka RISA)
Mathcad Prime	7.0.0.0	(aka MathCad)

5. ANALYSIS AND DESIGN PHILOSOPHY

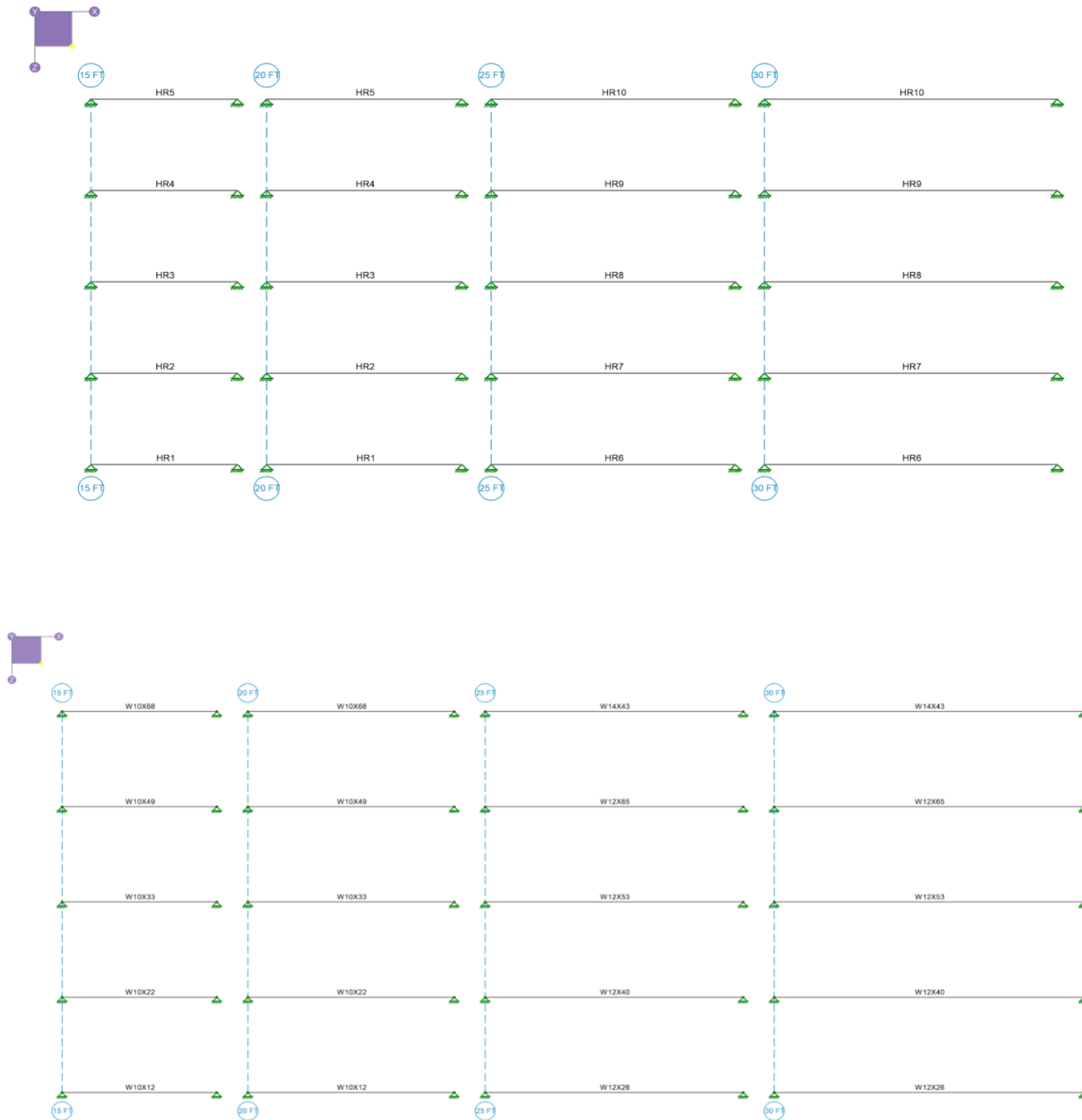
- The superstructure is to be conventional steel framing systems with a 1:4 ridge roof. Roof panels spec is to be per architectural drawings. Frame spacing dimensions are assumed to be from 15 FT to 30 FT, which would determined the span of the purlin.
- Purlins are to be hot rolled sections, spaced roughly at around 5 ft (1.524m) apart with no sag rods, designed as simply supported between frames with the effective length in compression of top flange equal to 0.5 in lieu of lateral supported provided to top flange by the roof panels.
- Purlins will be exposed to dead load (self-weight), Snow, Wind and Live loadings, we will determine the magnitude of each loading acting on the tributary area of each purlin. The required purlin section will likely be governed by frame spacing, strong axis bending and weak-axis bending, as well as max. deflection requirements
- Wind loading transferred into the purlins will be a function of outside pressure as well as internal suction. Total wind pressure is determined using 2015 NBC Structural commentary I static procedure considering worst-case exterior roof geometry.
- For simplicity, we will model the varies purlin sections at different lengths into RISA-3D (RISA) and use the results from the RISA model as part of the code an deflection check
- Due to the purlin steels being located at an angle on the roof, the horizontal component of the load that is parallel to the roof doesn't go through the shear center of the purlin. Therefore, roof purlins are assumed to be subjected to both biaxial bending and torsion. A hand check on torsion will be included near the end of this analysis.
- For simplicity, we will use a flexural analogy between torsion and ordinary bending for the torsion design, using methods outlined in Steel Structures Design And Behavior, Salmon, Johnson & Malhas (5th Ed), AISC Design Guide 9 & Limited States Design In Structural Steel, Kulak & Gilmor (8th Ed).
- The Flexural analogy is considered more conservative, as it considers axial compressive stress resulting from warping lateral flange bending as the most important, while shear stress component resulting from the wrapping torsion moment is less significant. This is best suited for a simple design in place of a time consuming torsion analysis, involving differential equations. We will also be applying a correction factor to the bending normal stresses from warping to bring in closer to results may have been from an elastic method.
- Since the Torsion Calculator from JDB 2.F.06 Torsion was made available in time of this calculation, we will use it for torsion check on the remaining member sizes outside of the Torsion Hand Check calculation.
- Since CISC CSA S16-01 Clause 14.10.4 only mentions that the maximum combined normal stress due to specified loads does not exceed F_y and doesn't specify a clear allowance limit. We will instead reference Section H2 of AISC LRFD standard which specs the combined normal stress for the factored loads does not exceed $0.9F_y$.

5.1. Selected WF section for RISA analysis

- Preliminary sizes were selected assuming 15' / 20' spans are no less than W10 and 25' / 30' spans are no less than W12. The respected 10 WF sections were selected for the analysis from the lighter section per their grouping based on the HSC blue book.

Hot Rolled Steel Section Sets

	Label	Shape	Type	Design List	Material	Design Rule	Area [in ²]	I _{yy} [in ⁴]	I _{zz} [in ⁴]	J [in ⁴]
1	HR1	W10X12	Beam	Wide Flange	A992	Typical	3.54	2.18	53.8	0.055
2	HR2	W10X22	Beam	Wide Flange	A992	Typical	6.49	11.4	118	0.239
3	HR3	W10X33	Beam	Wide Flange	A992	Typical	9.71	36.6	171	0.583
4	HR4	W10X49	Beam	Wide Flange	A992	Typical	14.4	93.4	272	1.39
5	HR5	W10X68	Beam	Wide Flange	A992	Typical	19.9	134	394	3.56
6	HR6	W12X26	Beam	Wide Flange	A992	Typical	7.65	17.3	204	0.3
7	HR7	W12X40	Beam	Wide Flange	A992	Typical	11.7	44.1	307	0.906
8	HR8	W12X53	Beam	Wide Flange	A992	Typical	15.6	95.8	425	1.58
9	HR9	W12X65	Beam	Wide Flange	A992	Typical	19.1	174	533	2.18
10	HR10	W14X43	Beam	Wide Flange	A992	Typical	12.6	45.2	428	1.05



6. LOAD DEVELOPMENT

Only selected load cases are considered applicable for purlin design, see Table 4-1 below. Seismic is not considered to govern due to relatively small proportion of dead load and calculation is for secondary members.

Basic Load Cases: The table below shows the basic load cases that were specified from the job data book Jdb2.20

Table 4-1 Basic Load Cases (BLCs)

Basic Load Case	Symbol	Description	Applied
BLC 1	SW	Self-weight, fireproofing, and misc. other dead load	Y
BLC 2	PO	Piping Operating Load	N
BLC 3	PE	Piping Empty Load	N
BLC 4	ET	Pipe / Equipment Test Load - "Content Weight Alone"	N
BLC 5	F	Pipe Friction Load	N
BLC 6	AT	Pipe Anchor Thermal	N
BLC 7	AF	Pipe Anchor Friction	N
BLC 8	AS	Pipe Anchor Seismic	N
BLC 9	GL	Guide Load	N
BLC 10	SG	Slug or Upset Loads	N
BLC 11	EL	Electrical	N
BLC 12	EO	Equipment Operating	N
BLC 13	EE	Equipment Empty	N
BLC 14	EM	Equipment Maintenance Load (ex. Bundle Pull)	N
BLC 15	L	Live Load	Y
BLC 16	S	Snow Load	Y
BLC 17	WT	Wind Transverse (Inward)	N
BLC 18	WL	Wind Longitudinal (Outward)	N
BLC 19	STOpe	Seismic Trans Operating	N
BLC 20	STEmp	Seismic Trans Empty	N
BLC 21	SLOpe	Seismic Long Operating	N
BLC 22	SLEmp	Seismic Long Empty	N
BLC 23	Di	Atmospheric Ice (Di), if applicable	N
BLC 24	TH,Stcon	Thermal Effects Contraction	N
BLC 25	TH,Stexp	Thermal Effects Expansion	N
BLC 26	-	Additional BLC, as required	-
Transportation Load Cases			
BLC 28	T1	Self-weight Longitudinal	N
BLC 29	T2	Self-weight Transverse	N
BLC 30	T3	Pipe Empty Longitudinal	N
BLC 31	T4	Pipe Empty Transverse	N
BLC 32	T5	Equipment Empty Longitudinal	N
BLC 33	T6	Equipment Empty Transverse	N
BLC 34	T7	Electrical Longitudinal	N
BLC 35	T8	Electrical Transverse	N
BLC 36	T9	Snow Longitudinal	N
BLC 37	T10	Snow Transverse	N
BLC 39	-	Additional BLC, as required	-
Enclosed Structure Wind - Include Internal Pressure or Suction			
BLC 40	WXPP	Building Wind Trans Positive Direction with Internal Pressure	Y
BLC 41	WZPP	Building Wind Long Positive Direction with Internal Pressure	N
BLC 42	WXNP	Building Wind Trans Negative Direction with Internal Pressure	Y

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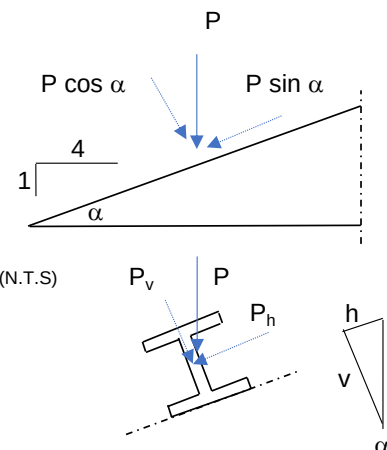
BLC 43	WZNP	Building Wind Long Negative Direction with Internal Pressure	N
BLC 44	WXPS	Building Wind Trans Positive Direction with Internal Suction	Y
BLC 45	WZPS	Building Wind Long Positive Direction with Internal Suction	N
BLC 46	WXNS	Building Wind Trans Negative Direction with Internal Suction	Y
BLC 47	WZNS	Building Wind Long Negative Direction with Internal Suction	N

Geometry - Based on assumed 1:4 roof slope

Roof Slope Angle $\alpha = \tan^{-1}(1/4) = 14.0$ Degrees

- Due to the purlin steels will be located at an angle on the roof, we will have to consider there are minor directional forces applied to all of the gravity related loadings below as well.

$P_v = P \cos \alpha = 0.970 P$ (Angle Adjustments)
 $P_h = P \sin \alpha = 0.243 P$



6.1 BLC 1 - Self-weight

Steel Self-weight will of the purlin including an additional 15% factor considered for connection weight

Design Steel Dead Load $SW_p = -1.15$ (Y-Axis, no angle)
 Design Steel Dead Load (Vert. direct. w/ angle adjusted) $SW_{pv} = -1.12$ (Risa Input)(Y-Axis)
 Design Steel Dead Load (Hori. direct. w/ angle adjusted) $SW_{ph} = 0.28$ (Risa Input)(Z-Axis)

6.1.1 Roof Self-weight

Roof type used on the project is expected to vary, so preliminary roof weight is based on roof type spec system R-1 used in BLDG-8A (240kV Substation).

ROOF TYPE/ASSEMBLY

R-1 - SR2 OR BETTER (BY VICWEST OR APPROVED EQUAL) PRE-INSULATED NONCOMBUSTIBLE METAL PANEL ROOF ASSEMBLY WITH 0.0336" THK (22G) PREFINISHED GALVANIZED METAL CLAD BOTH FACES R30 INSULATION INSIDE ROOF CAVITY WEATHER TIGHT, STANDING SEAM (2" DEEP MINIMUM) INSULATED METAL ROOF PANEL MECHANICALLY SEALED C/W DOUBLE LOCK.

Unit weight of roof panel (~4" assumed at 4 psf) $q_{roof} = 0.3$ kPa
 Unit weight of Electrical / Lighting (~3 psf) $q_{el} = 0.2$ kPa
 Unit weight of HVAC Ducting, Heater / Fan (~4 psf) $q_{HVAC} = 0.3$ kPa
 Unit weight of Misc (attachments) $q_{MISC} = 0.2$ kPa

Assumed roof dead load (sum of above) $q_p = 1.00$ kPa
 (Inline with Dow FSZ-0001-01 Civil Engineering Design Criteria Min. Dead Load)

Design Roof Dead Load (~ 5 FT trib. span) $W_p = 1.52$ kN/m
 Design Roof Dead Load (Vert. direct. w/ angle adjusted) $W_{pv} = 1.48$ kN/m (Risa Input)
 Design Roof Dead Load (Hori. direct. w/ angle adjusted) $W_{ph} = 0.37$ kN/m (Risa Input)

6.2 BLC15 : Live Load (LL)

Per Dow FSZ-0001-01 Civil Engineering Design Criteria to use local design code for live load - NBC 2019 (AE)

Roof Live Load (Min Area Load)(Table 4.1.5.3)	or	1.00 kPa	Gov't
Roof Live Load (Min. Concentrated Load)(Table 4.1.5.9)		1.30 kN	
Design Roof Live Load (~ 5 FT trib. span)	L_p	= 1.52 kN/m	
Design Roof Live Load (Vert. direct. w/ angle adjusted)	L_{pv}	= 1.48 kN/m	(Risa Input)
Design Roof Live Load (Hori. direct. w/ angle adjusted)	L_{ph}	= 0.37 kN/m	(Risa Input)

6.3 BLC16 : Snow Load (S)

Per Dow FSZ-0001-01 Civil Engineering Design Criteria to use local design code for snow load - NBC 2019 (AE)

Snow Load Parameters

$$S = I_s [S_s (C_b C_w C_s C_a) + S_r]$$

$I_s = 1$	Importance Factor	normal	for ULS	As per - NBC 2019 AE 4.1.6.2.
$S_s = 1.6$	1/50 year ground snow load kPa			Normal considered for secondary mem.
$C_b = 0.8$	Basic roof snow load factor			Per Dow FS2Z-0001-33, Section 4.6
$C_w = 1$	Wind exposure factor			Per Dow FS2Z-0001-33, Section 4.6
$C_s = 1$	Slope factor			NBC 2019 AE 4.1.6.2 - Sentence (4)
$C_a = 1$	Shape factor			<30% Slope - NBC 2019 AE 4.1.6.2
$S_r = 0.1$	1/50 year associated rain load			- Sentence (5)
				As per Dow FS2Z-0001-33, Section 4.6

ULS $S = 1.38 \text{ kPa}$

Risa factor for SLS = $0.9 / 1.0 = 0.9$

NBC 2019 AE, Table 4.1.6.2 A.

Snow Load on Roof Beams:

Design Roof Snow Load (~ 5 FT trib. span)	S_p	= 2.1 kN/m	
Design Roof Snow Load (Vert. direct. w/ angle adjusted)	S_{pv}	= 2.0 kN/m	(Risa Input)
Design Roof Snow Load (Hori. direct. w/ angle adjusted)	S_{ph}	= 0.5 kN/m	(Risa Input)

6.4 BLC 17 - Wind Loading

Wind load calculations are as per NBC (AE) 2019. Reference calculation site parameters are taken from Doc. FS2Z-0001-33. Reference wind band height taken as 35' (10.67 m) based on the tallest anticipated building. Building height is assumed to be < 20 m and less than the smaller plan dimension. No angle adjustment required as the wind loading is assumed to be perpendicular to the roof surface

6.4.1 Wind Loading Exterior Pressure / Suction

Tri. Area (m²) Calc: Typ trib. sp. of ~5' (1.524 m)

Cols spacing ranges from 15' (4.572 m), 20' (6.096 m), 25' (7.62 m) to 30' (9.144 m)

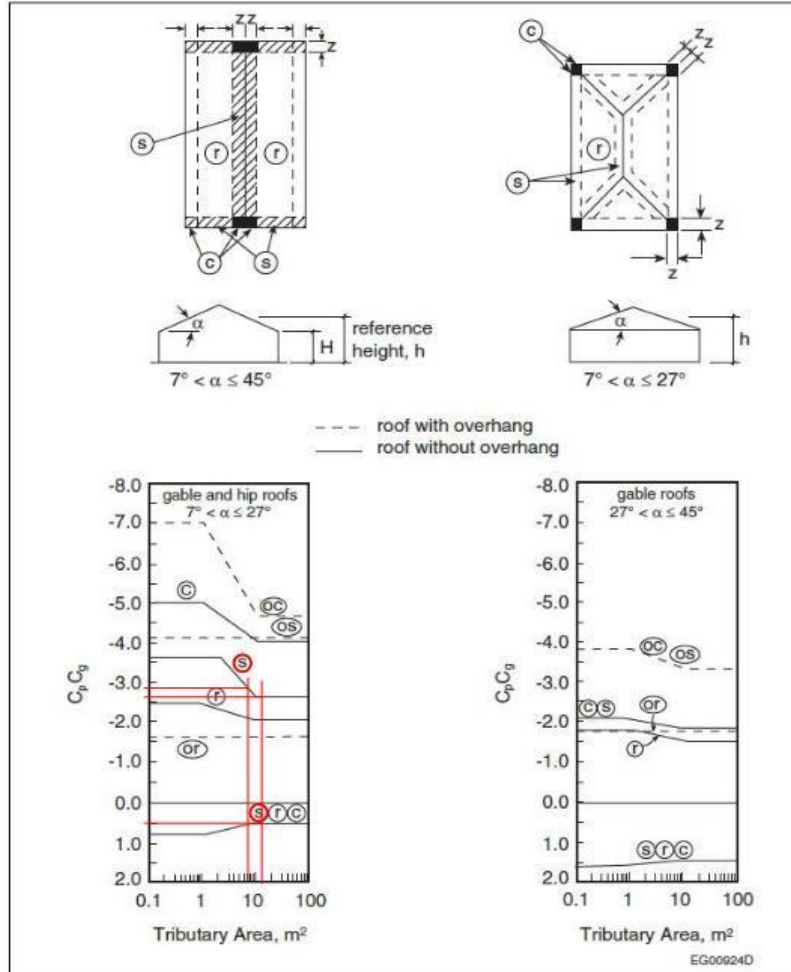
-Since the tributary area depends on the frame / column spacing, which is on a range of values, we will use the worse case that produces the largest CpCg for simplicity.

$$A_t = 4.572 \times 1.524 = 6.968 \text{ m}^2$$

$$A_t' = 9.144 \times 1.524 = 13.94 \text{ m}^2$$

- Also we will assume S zone values for worse case

Figure 4.1.7.6.-E
External peak values of $C_p C_g$ on single-span gabled and hipped roofs with a slope greater than 7°
for the design of structural components and cladding
Forming Part of Sentence 4.1.7.6.(6)



Notes to Figure 4.1.7.6.-E:

- (1) Coefficients for overhanging roofs have the prefix "o" and refer to the same roof areas as referred to by the corresponding symbol without a prefix. They include contributions from both upper and lower surfaces.
- (2) End-zone width z is the lesser of 10% of the least horizontal dimension and 40% of height, H , but not less than 4% of the least horizontal dimension or 1 m.
- (3) Combinations of external and internal pressures must be evaluated to obtain the most severe loading.
- (4) Positive coefficients denote forces towards the surface, whereas negative coefficients denote forces away from the surface. Each structural element must be designed to withstand forces of both signs.
- (5) For hipped roofs with $7^\circ < \alpha \leq 27^\circ$, edge/ridge strips and pressure-gust coefficients for ridges of gabled roofs apply along each hip.

$C_p C_g$ Values:

+ve - towards surface
-ve - away from surface

ULS wind importance factor
Reference velocity pressure
Reference height
Open terrain exposure factor, varies by height
Topographic factor
External peak values of $C_p C_g$ Acting Inwards
External peak values of $C_p C_g$ Acting Outwards
ULS wind pressure for open structure

I_w = 1 Normal considered for secondary member
 q = 0.43 kPa Per Dow FS2Z-0001-33, Section 4.7.1
 h = 10.67 m
 C_e = 1.013 $(h/10)^{0.2}$
 C_t = 1 NBC 2019 AE, Table 4.1.7.4
 $C_p C_{g,in}$ = 0.5 NBC 2019 AE, Table 4.1.7.6 E
 $C_p C_{g,out}$ = -2.8 NBC 2019 AE, Table 4.1.7.6 E
 p = $I_w q C_e C_t C_g C_p$ (Wind Pressure EQ - 2019 NBC, AE, 4.1.7.3.)

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ULS Exterior wind pressure acting Inwards	$p_{ext,in}$	=	0.22	kPa
ULS Exterior wind pressure acting Outwards	$p_{ext,out}$	=	-1.22	kPa
Ext. Wind Load Acting Inwards (~ 5 FT trib. span)	$w_{WL,ext,in}$	=	0.33	kN/m (WXP)
Ext. Wind Load Acting Outwards (~ 5 FT trib. span)	$w_{WL,ext,out}$	=	-1.86	kN/m (WXN)

6.4.2 Wind Loading Interior Pressure / Suction

4.1.7.7. Internal Pressure Coefficient

1) The internal pressure coefficient, C_{pi} , shall be as prescribed in Table 4.1.7.7.

Table 4.1.7.7.
Internal Pressure Coefficients
 Forming Part of Sentence 4.1.7.7.(1)

Building Openings	Values for C_{pi}
Uniformly distributed small openings amounting to less than 0.1% of the total surface area of the building	-0.15 to 0.0
Non-uniformly distributed openings of which none is significant or significant openings that are wind-resistant and closed during storms	-0.45 to +0.30
Large openings likely to remain open during storms	-0.70 to +0.70

ULS wind importance factor	I_w	=	1	+ve - towards surface -ve - away from surface
Reference velocity pressure	q	=	0.43	kPa
Open terrain exposure factor, varies by height	C_e	=	1.013	$(h/10)^{0.2}$
Topographic factor	C_t	=	1	
Internal Gust Effect Factor	C_g	=	2	(From 4.1.7.3. Sentence 8 & 10)
Internal Pressure Coefficient Acting Inwards	$C_{p,int,Pressure}$	=	-0.45	(From Table 4.1.7.7., assume non-
Internal Pressure Coefficient Acting Outwards	$C_{p,int,suction}$	=	0.3	uniformly distributed openings)
Reference height	h	=	10.67	m
ULS wind pressure for open structure	p	=	$I_w q C_e C_t C_g C_p$	
ULS Interior wind pressure acting Inwards	$p_{int,in. p}$	=	-0.39	kPa
ULS Interior wind pressure acting Outwards	$p_{int,out. s}$	=	0.26	kPa

Design wind load acting on the purlin is a function of purlin spacing. Assume 5' (1.524 m)

Int. Wind Load Pressure (~ 5 FT trib. span)	$w_{WL,int,pressure}$	=	-0.60	kN/m (Pressure)
Int. Wind Load Suction (~ 5 FT trib. span)	$w_{WL,int,suction}$	=	0.40	kN/m (Suction)

6.4.3 Total Design Wind Loading

-For simplicity, we will assume wind loadings are created from external transverse wind in the X-Axis (WX), which creates both inward or outward wind loading (P/N), resulting interior pressure / suction (P/S) acting on the purlins.

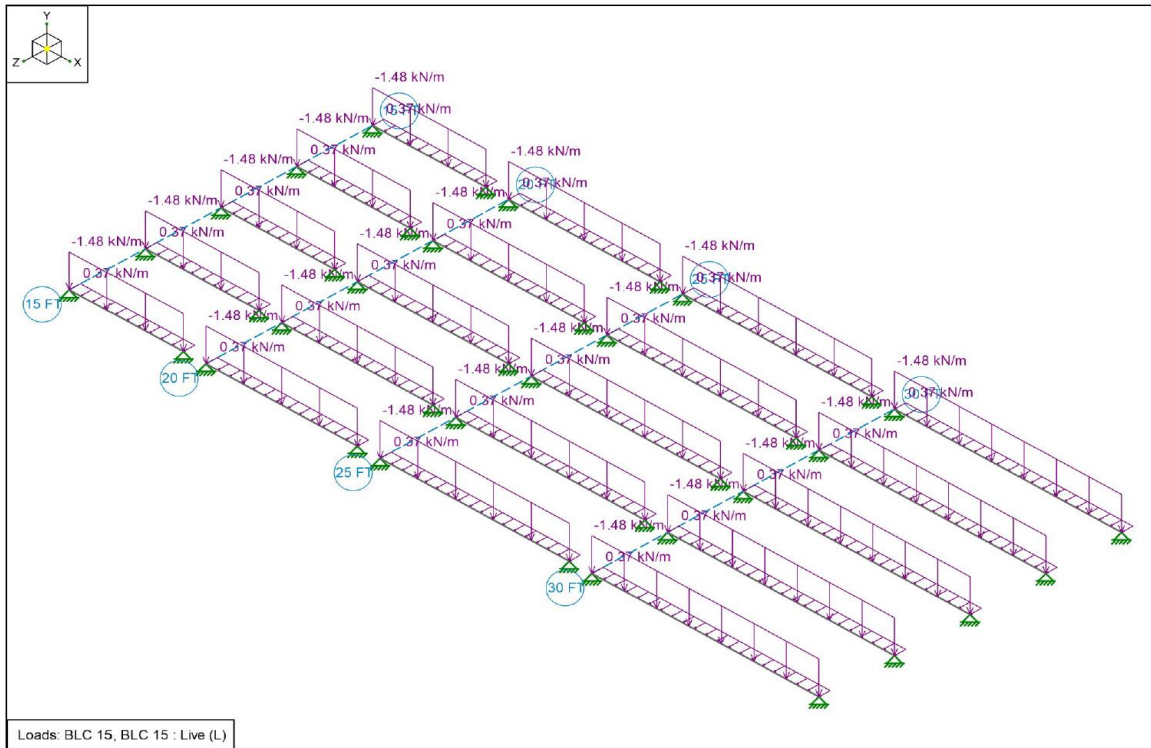
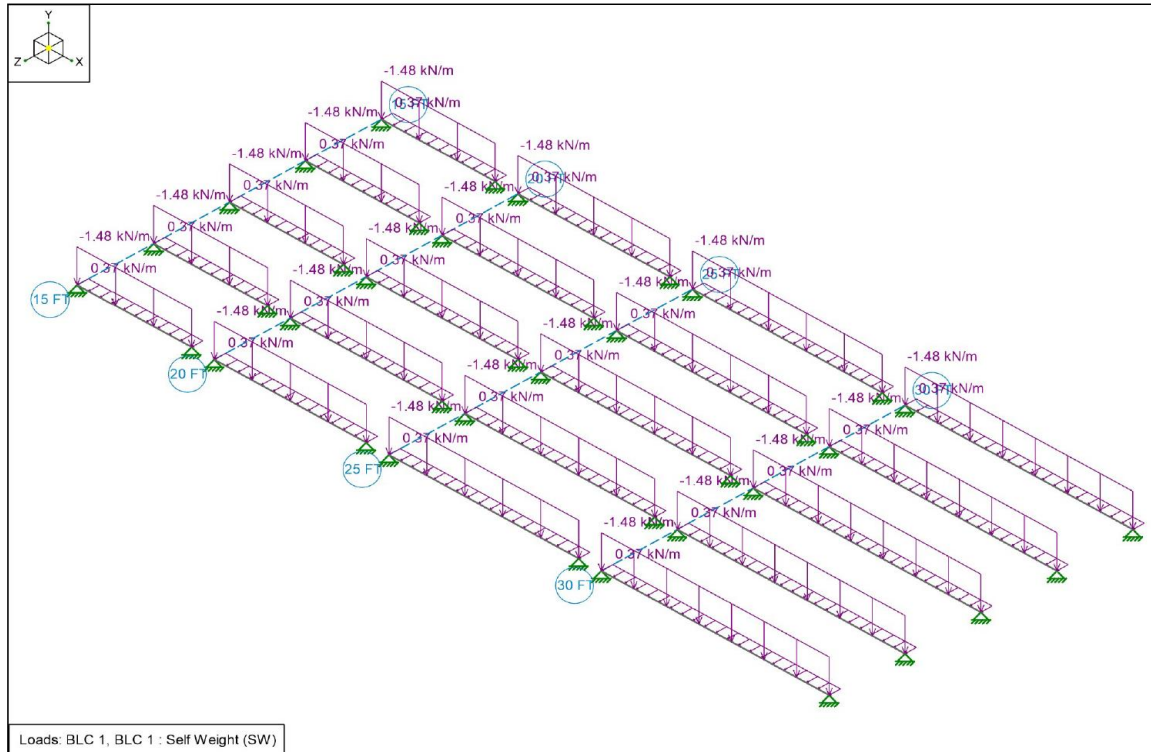
-Therefore, we will have to include the related four wind load cases.

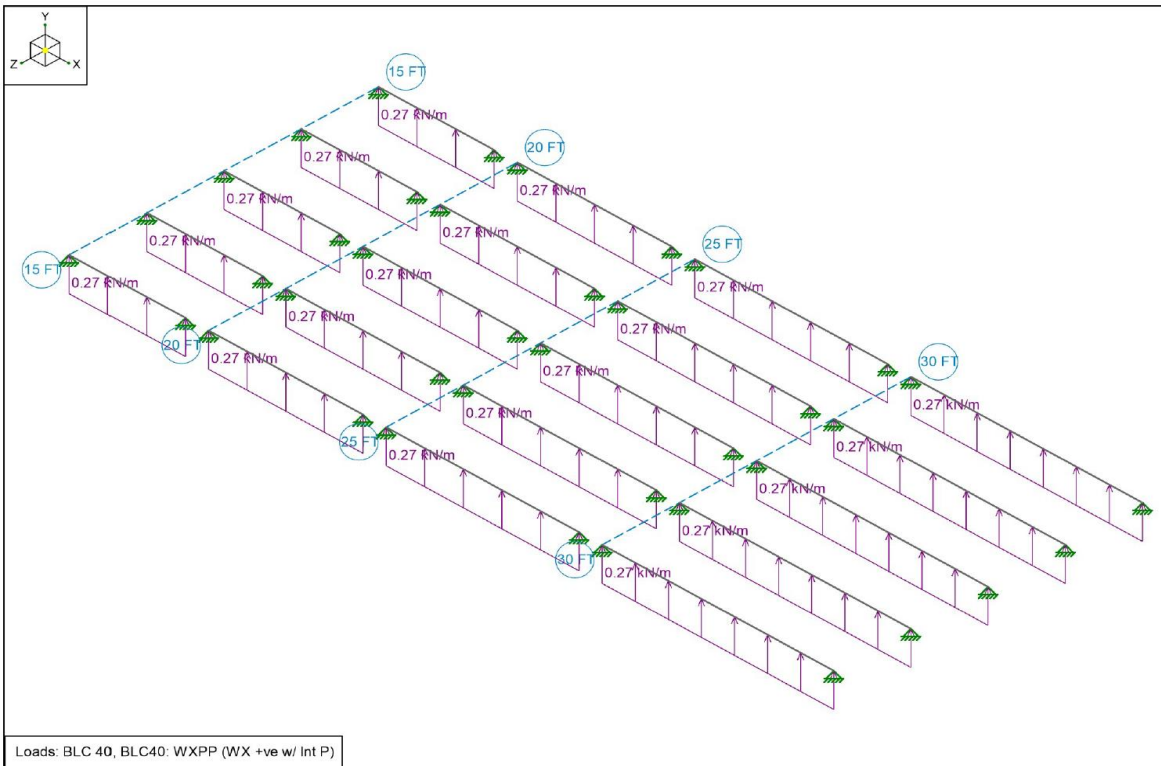
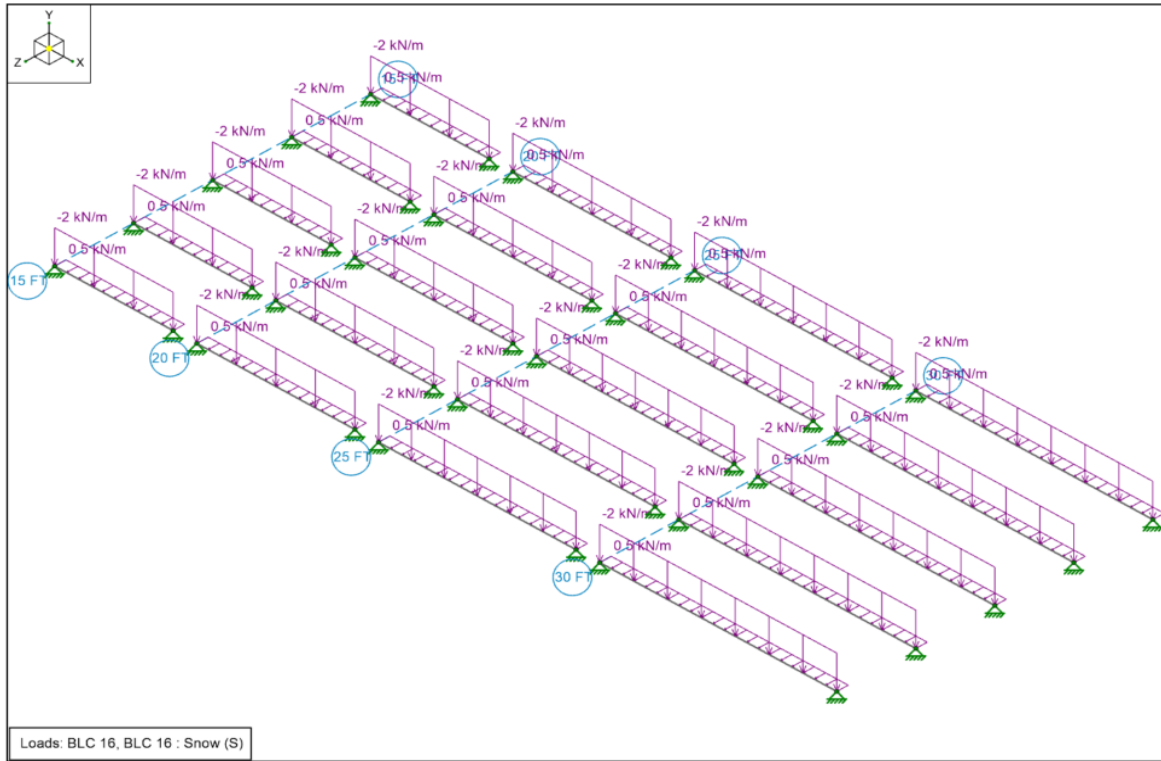
BLC40: WXPP (Bldg Wind Trans +ve Dir. w/ Int. Press.)	$w_{WLT, WXPP}$	=	0.33+-0.6	=	-0.27	kN/m
BLC42: WXNP (Bldg Wind Trans -ve Dir. w/ Int. Press.)	$w_{WLT, WXNP}$	=	-1.86+-0.6	=	-2.46	kN/m (Risa
BLC44: WXPS (Bldg Wind Trans +ve Dir. w/ Int. Suction)	$w_{WLT, WXPS}$	=	0.33+0.4	=	0.73	kN/m Input)
BLC46: WXNS (Bldg Wind Trans -ve Dir. w/ Int. Suction)	$w_{WLT, WXNS}$	=	-1.86+0.4	=	-1.46	kN/m

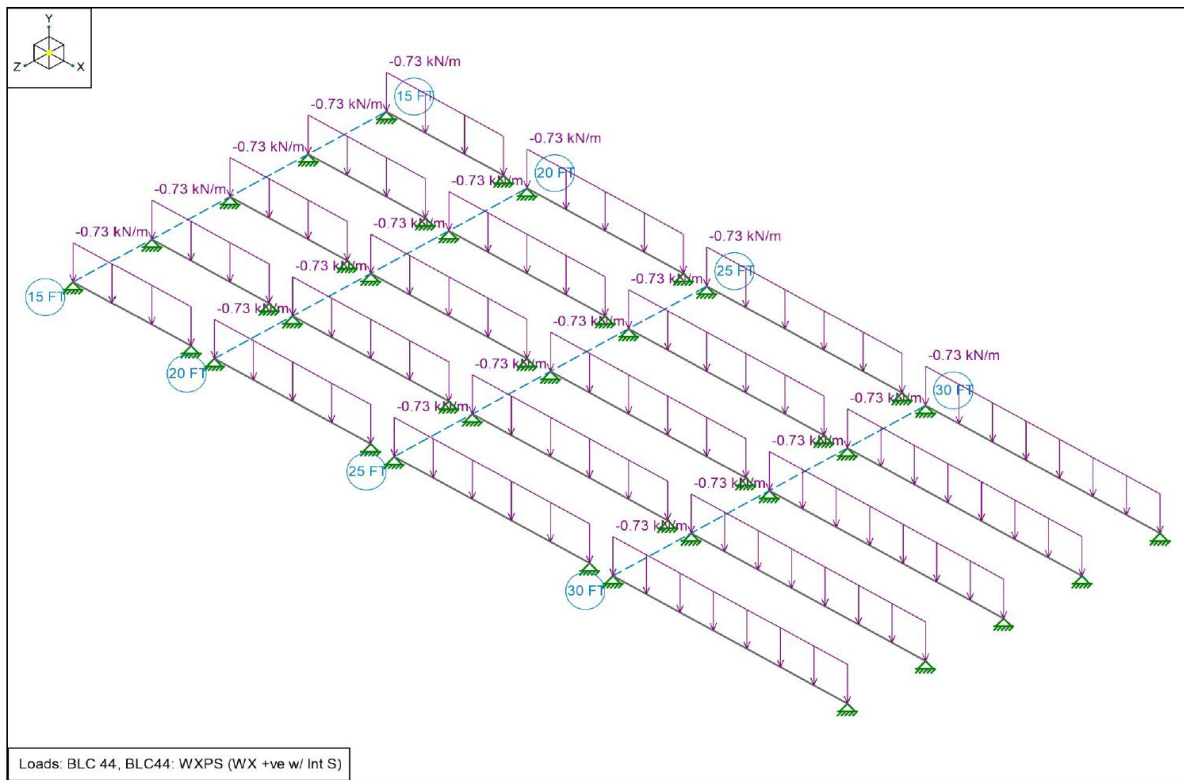
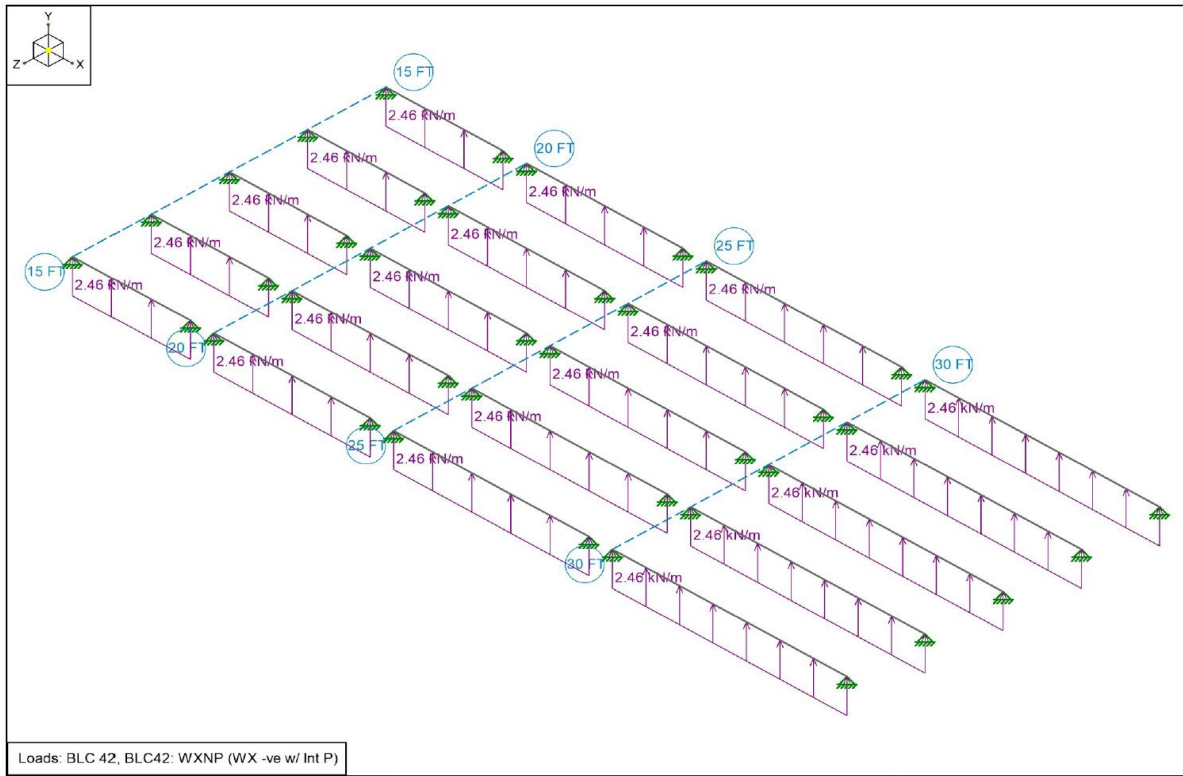
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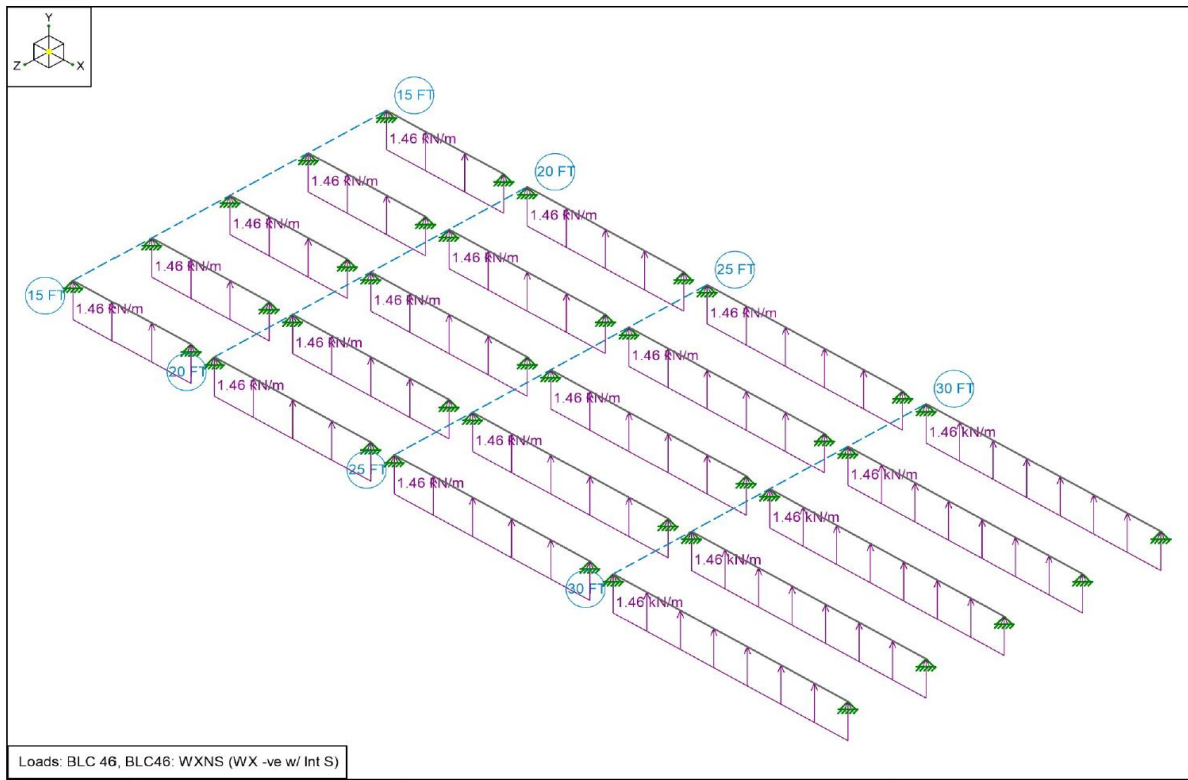
Project No: P2ZU

6.5 RISA Inputs









7. LOAD COMBINATIONS

Applicable load combinations for Purlin design include the following:

	<u>Principal Loads</u>	<u>Companion Loads</u>
1	1.4DLO	
2	1.25DLO + 1.5LL	1.0S or 0.4W
3	1.25DLO + 1.4WL	0.5L or 0.5S

- Due to the angle on the purlin, Dead, Live & Snow loads will induce weak-axis bending, while wind loading will induce strong-axis bending. The governing loading scenario will include bi-axial bending and bi-axial shear.

- Refer to JDB 2.B.04 and attachment 1 -Buildings BLC-LCs.xlsx for the full set of combinations.

- The steel superstructure of the building is to be stick built on site and therefore combinations related to land transport, lifting analysis and crane will be excluded.

- Usually only serviceability Load Combinations (SLS) - LC 372 to LC 478, should be used for the deflection checks. However, for simplicity and ease, we will include envelop results from all combinations for this calculation

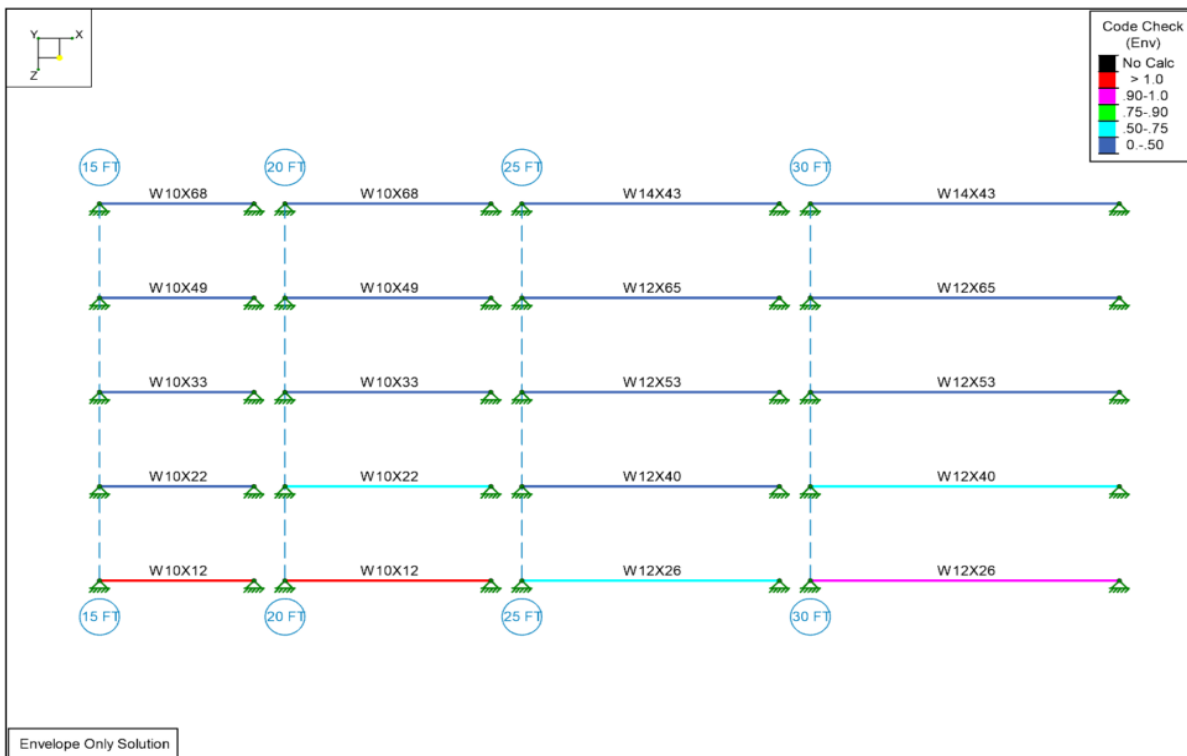
8. ANALYSIS AND RESULTS

Analysis were preformed by RISA where the purlin shear and bending capacity were checked to CSA S16-14. The purlins experienced bi-axial bending and shear as well as local bending about the web caused by eccentric dead load.

8.1 Member Utility

8.1.1 Unity Check

- Failed and high unity members were left in to show that they are limitation for using lighter members.
- However, we should also avoid using the lightest sections, especially for long span members since the higher utility are a result of estimated loadings.



8.2 Deflection Checks

- Per Dow spec, max deflection for roof purlins is $L/200$ and should be determined using total loads without factors or reduction, excluding concentrated load. Therefore, only deflection results from the Serviceability Load Combinations (SLS) - LC 372 to LC 478, should count for the deflection limit check.
- However, for simplicity and ease, we will include envelop results from all combinations for this calculation. (Conservative)

8.2.1 Member Deflection Limits

L/200	Spans @ 15'		Spans @ 20'		Spans @ 25'		Spans @ 30'	
	In	mm	In	mm	In	mm	In	mm
Dmax	0.90	22.9	1.20	30.5	1.50	38.1	1.80	45.7

8.2.2 RISA Member Deflection Envelope Results

Envelope Beam Deflections

Member Label	Span	Location [ft]	y' [mm]	(n) L/y' Ratio	LC
1 P1	1	max 0.156	0.026	NC	310
2	1	min 7.5	-8.356	547	88
3 P2	1	max 1.719	-0.026	NC	152
4	1	min 7.5	-3.929	1163	88
5 P3	1	max 14.688	-0.025	NC	555
6	1	min 7.5	-2.8	1632	88
7 P4	1	max 11.406	-0.025	NC	310
8	1	min 7.5	-1.843	2481	88
9 P5	1	max 14.688	-0.026	NC	477
10	1	min 7.5	-1.338	3416	88
11 P6	1	max 1.875	0.028	NC	152
12	1	min 10	-26.409	230	88
13 P7	1	max 0.625	0.026	NC	297
14	1	min 10	-12.416	491	88
15 P8	1	max 1.667	-0.027	NC	297
16	1	min 10	-8.851	688	88
17 P9	1	max 19.583	-0.025	NC	581
18	1	min 10	-5.823	1046	88
19 P10	1	max 0.208	-0.025	NC	318
20	1	min 10	-4.23	1441	88
21 P11	1	max 23.698	0.027	NC	297
22	1	min 12.5	-17.743	429	88
23 P12	1	max 21.875	0.027	NC	310
24	1	min 12.5	-12.274	620	88
25 P13	1	max 22.396	-0.028	NC	144
26	1	min 12.5	-9.202	828	88
27 P14	1	max 0.781	0.029	NC	151
28	1	min 12.5	-7.579	1005	88
29 P15	1	max 9.635	-0.025	NC	310
30	1	min 12.5	-8.881	858	88
31 P16	1	max 0.313	0.032	NC	577
32	1	min 15	-36.791	248	88
33 P17	1	max 0.313	-0.026	NC	297
34	1	min 15	-25.451	359	88
35 P18	1	max 1.563	-0.029	NC	144
36	1	min 15	-19.082	479	88
37 P19	1	max 0.313	-0.032	NC	310
38	1	min 15	-15.715	581	88
39 P20	1	max 25.313	-0.027	NC	310
40	1	min 15	-18.415	496	88

- By inspection, member deflections are below the limits for their respective lengths, even under factored load combinations. Therefore, we will conclude that the member selection is sufficient for deflection as in reality the deflection may even be less as the roof panel would provide an additional level of stiffness.

8.3 Connection Design (In-House)

- Assuming the scope is done by 3rd party and not part of the scope for this analysis.

8.4 Connection Design Tables (For Fabricators)

8.4.1 RISA Reaction Envelope Results

Envelope Node Reactions

Node Label	X [kN]	LC	Y [kN]	LC	Z [kN]	LC	MX [kN-m]	LC	MY [kN-m]	LC	MZ [kN-m]	LC
1 N11	max 0	612	20.044	269	0	586	0	612	0	612	0	612
2	min 0	70	-5.897	131	-5.011	88	0	70	0	70	0	70
3 N22	max 0	612	26.144	269	0	586	LOCKED	0	612	0	612	0
4	min 0	70	-6.588	131	-6.536	88	LOCKED	0	70	0	70	0
5 N31	max 0	612	31.372	269	0	586	0	612	0	612	0	612
6	min 0	70	-7.905	131	-7.843	88	0	70	0	70	0	70
7 N21	max 0	612	26.144	269	0	586	0	612	0	612	0	612
8	min 0	70	-6.588	131	-6.536	88	0	70	0	70	0	70
9 N1	max 0	612	15.033	269	0	586	LOCKED	0	612	0	612	0
10	min 0	70	-4.423	131	-3.758	88	LOCKED	0	70	0	70	0
11 N12	max 0	612	20.044	269	0	586	LOCKED	0	612	0	612	0

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12		min	0	70	-5.897	131	-5.011	88	LOCKED		0	70	0	70
13	N2	max	0	612	15.033	269	0	586	0	612	0	612	0	612
14		min	0	70	-4.423	131	-3.758	88	0	70	0	70	0	70
15	N32	max	0	612	31.372	269	0	586	LOCKED		0	612	0	612
16		min	0	70	-7.905	131	-7.843	88	LOCKED		0	70	0	70
17	N3	max	0	612	15.502	269	0	586	0	612	0	612	0	612
18		min	0	70	-4.085	131	-3.875	88	0	70	0	70	0	70
19	N4	max	0	612	15.502	269	0	586	LOCKED		0	612	0	612
20		min	0	70	-4.085	131	-3.875	88	LOCKED		0	70	0	70
21	N13	max	0	612	20.669	269	0	586	0	612	0	612	0	612
22		min	0	70	-5.447	131	-5.167	88	0	70	0	70	0	70
23	N14	max	0	612	20.669	269	0	586	LOCKED		0	612	0	612
24		min	0	70	-5.447	131	-5.167	88	LOCKED		0	70	0	70
25	N23	max	0	612	27.216	269	0	586	0	612	0	612	0	612
26		min	0	70	-6.186	579	-6.804	88	0	70	0	70	0	70
27	N24	max	0	612	27.216	269	0	586	LOCKED		0	612	0	612
28		min	0	70	-6.186	579	-6.804	88	LOCKED		0	70	0	70
29	N33	max	0	612	32.66	269	0	586	0	612	0	612	0	612
30		min	0	70	-7.423	579	-8.165	88	0	70	0	70	0	70
31	N34	max	0	612	32.66	269	0	586	LOCKED		0	612	0	612
32		min	0	70	-7.423	579	-8.165	88	LOCKED		0	70	0	70
33	N5	max	0	612	16.014	269	0	586	0	612	0	612	0	612
34		min	0	70	-3.717	131	-4.003	88	0	70	0	70	0	70
35	N6	max	0	612	16.014	269	0	586	LOCKED		0	612	0	612
36		min	0	70	-3.717	131	-4.003	88	LOCKED		0	70	0	70
37	N15	max	0	612	21.351	269	0	586	0	612	0	612	0	612
38		min	0	70	-4.956	131	-5.338	88	0	70	0	70	0	70
39	N16	max	0	612	21.351	269	0	586	LOCKED		0	612	0	612
40		min	0	70	-4.956	131	-5.338	88	LOCKED		0	70	0	70
41	N25	max	0	612	28.25	269	0	586	0	612	0	612	0	612
42		min	0	70	-6.186	579	-7.062	88	0	70	0	70	0	70
43	N26	max	0	612	28.25	269	0	586	LOCKED		0	612	0	612
44		min	0	70	-6.186	579	-7.062	88	LOCKED		0	70	0	70
45	N35	max	0	612	33.899	269	0	586	0	612	0	612	0	612
46		min	0	70	-7.423	579	-8.475	88	0	70	0	70	0	70
47	N36	max	0	612	33.899	269	0	586	LOCKED		0	612	0	612
48		min	0	70	-7.423	579	-8.475	88	LOCKED		0	70	0	70
49	N7	max	0	612	16.759	269	0	586	0	612	0	612	0	612
50		min	0	70	-3.712	579	-4.19	88	0	70	0	70	0	70
51	N8	max	0	612	16.759	269	0	586	LOCKED		0	612	0	612
52		min	0	70	-3.712	579	-4.19	88	LOCKED		0	70	0	70
53	N17	max	0	612	22.345	269	0	586	0	612	0	612	0	612
54		min	0	70	-4.949	579	-5.586	88	0	70	0	70	0	70
55	N18	max	0	612	22.345	269	0	586	LOCKED		0	612	0	612
56		min	0	70	-4.949	579	-5.586	88	LOCKED		0	70	0	70
57	N27	max	0	612	29.177	269	0	586	0	612	0	612	0	612

Envelope Node Reactions (Continued)

Node Label	X [kN]	LC	Y [kN]	LC	Z [kN]	LC	MX [kN-m]	LC	MY [kN-m]	LC	MZ [kN-m]	LC
58		min	0	70	-6.186	579	-7.294	88	0	70	0	70
59	N28	max	0	612	29.177	269	0	586	LOCKED		0	612
60		min	0	70	-6.186	579	-7.294	88	LOCKED		0	70
61	N37	max	0	612	35.012	269	0	586	0	612	0	612
62		min	0	70	-7.423	579	-8.753	88	0	70	0	70
63	N38	max	0	612	35.012	269	0	586	LOCKED		0	612
64		min	0	70	-7.423	579	-8.753	88	LOCKED		0	70
65	N9	max	0	612	17.633	269	0	586	0	612	0	612
66		min	0	70	-3.712	579	-4.408	88	0	70	0	70
67	N10	max	0	612	17.633	269	0	586	LOCKED		0	612
68		min	0	70	-3.712	579	-4.408	88	LOCKED		0	70
69	N19	max	0	612	23.511	269	0	586	0	612	0	612
70		min	0	70	-4.949	579	-5.878	88	0	70	0	70
71	N20	max	0	612	23.511	269	0	586	LOCKED		0	612
72		min	0	70	-4.949	579	-5.878	88	LOCKED		0	70
73	N29	max	0	612	27.455	269	0	586	0	612	0	612
74		min	0	70	-6.186	579	-6.864	88	0	70	0	70
75	N30	max	0	612	27.455	269	0	586	LOCKED		0	612
76		min	0	70	-6.186	579	-6.864	88	LOCKED		0	70
77	N39	max	0	612	32.946	269	0	586	0	612	0	612
78		min	0	70	-7.423	579	-8.236	88	0	70	0	70
79	N40	max	0	612	32.946	269	0	586	LOCKED		0	612
80		min	0	70	-7.423	579	-8.236	88	LOCKED		0	70
81	Totals:	max	0	612	985.983	269	0	586				
82		min	0	70	-222.693	579	-246.496	88				

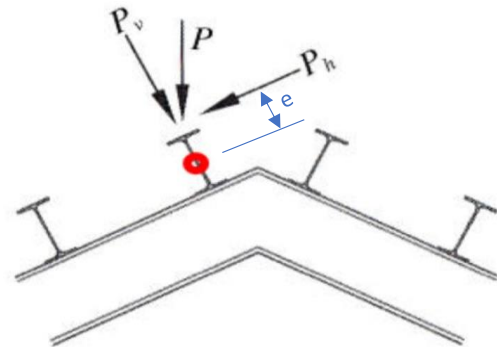
8.4.2 Table For Connection Design (For Fabricators)

- Following table is developed from RISA model max reaction values, including factored combinations, rounded to ~1.1x and nearest 0.5 for simplicity.
- Connections are assumed to be pinned, therefore maximum connection forces consider maximum vertical and horizontal shear loads. Vertical shear (Vy) is caused by self-weight while Horizontal shear (Vh) is caused by live, snow & wind loading.

Design Forces	Spans @ 15'		Spans @ 20'		Spans @ 25'		Spans @ 30'	
	V _{y, 15'} (KN)	V _{h, 15'} (KN)	V _{y, 20'} (KN)	V _{h, 20'} (KN)	V _{y, 25'} (KN)	V _{h, 25'} (KN)	V _{y, 30'} (KN)	V _{h, 30'} (KN)
Max	19.5	5.0	26	6.5	32.5	8.5	39	10

8.5 Torsion Hand Check

- By observation, critical section is P17 - W12x40 @30' with critical load comb. from the envelop code check being LC 269, which consists mostly of 1.25DLO+1.5SL+1.0 LL.
- From RISA model Loadings applied for LC269, we know that factored loadings are equal to the following, we'll add the self weight to the critical section in both directions since that was automatically calculated in RISA but required for the hand check.



$$\begin{aligned}
 \text{Design Roof Ld (Vert. dir.) } W'_{Pv} &= 6.33 + 1.25 * 0.58 * 0.970 = 7.03 \text{ kN/m (Spread Input)} \\
 \text{Design Roof Ld (Hori. dir.) } W'_{Ph} &= 1.582 + 1.25 * 0.58 * 0.243 = 1.76 \text{ kN/m (Spread Input)}
 \end{aligned}$$

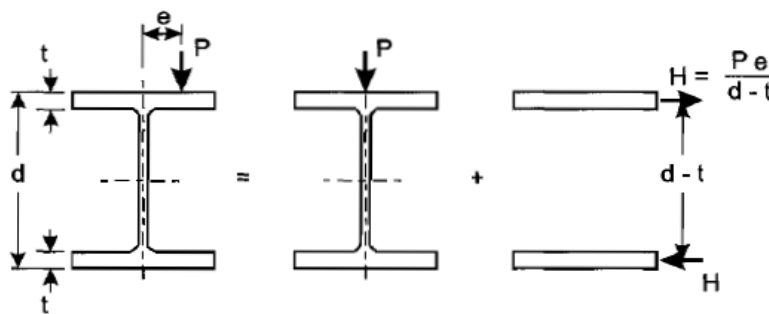
↑ Risa Loading
↑ DL SF
↑ W12x40 SW
↑ Angle Adj.

Eccentric (e)

- Assumed e distance is from shear center to the center of panel, which is equal to full beam depth plus the assumed roof panel thickness of 4" divided the total by 2

$$e = (11.9 + 4) / 2 = 7.95 \text{ in or } 0.202 \text{ m}$$

Flexural Analogy

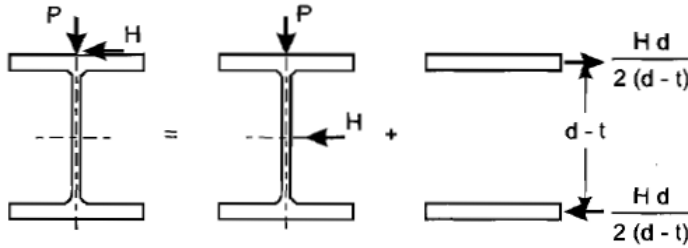


(a) Eccentric vertical load

Fig 5.16 - Limited States Design In Structural Steel, Kulak & Gilmor (8th Ed)

Section W12x40 (W310x60)

Class	1
d	= 303 mm
b	= 203 mm
t	= 13.1 mm
w _d	= 0.580 kN/m
S _x	= 842 x 10 ³ mm ³
Z _x	= 933 x 10 ³ mm ³
S _y	= 180 x 10 ³ mm ³
Z _y	= 275 x 10 ³ mm ³
J	= 378 x 10 ³ mm ⁴
C _w	= 384 x 10 ⁹ mm ⁶



(b) Vertical and horizontal loads

Figure 5.16 – Flexure Analogy

$$\begin{aligned} G &= 77000 \text{ MPa} \\ E &= 200000 \text{ MPa} \\ F_y &= 350 \text{ MPa} \\ F_u &= 450 \text{ MPa} \end{aligned}$$

$$\begin{aligned} L &= 9.144 \text{ m} \\ h &= d - t \\ h &= 0.29 \text{ m} \end{aligned}$$

$$\begin{aligned} \mu &= \text{SQRT}(GJ/EC_w) \\ &= 0.0006 \\ \mu L &= 5.63 \end{aligned}$$

Total Stress = Strong Axis Bending + Weak Axis Bending + Warping * (β)

Strong Axis Bending & Stress

$$M_x = w_v * L^2 / 8 = 7.03 * (9.14)^2 / 8 = 73.5 \text{ kNm}$$

$$f_x = M_x / S_x = 73.5 / 842 * 1000 = 87.3 \text{ MPa}$$

Weak Axis Bending & Stress

$$M_y = w_h * L^2 / 8 = 1.76 * (9.14)^2 / 8 = 18.4 \text{ kNm}$$

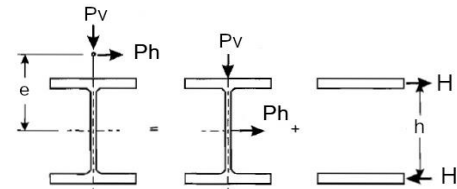
$$f_y = M_y / S_y = 18.4 / 180 * 1000 = 102.2 \text{ MPa}$$

Torsion (Created by P_H)

$$T = w_h * e = 1.76 * 0.202 = 0.36 \text{ kNm/m}$$

Couple Force Equivalent (Created by Torsion conversion)

$$H = T / h = 0.36 / 0.29 = 1.2 \text{ kNm}$$



Warping & Stress

$$M_f = H * L^2 / 8 = 1.24 * (9.14)^2 / 8 = 13.0 \text{ kNm}$$

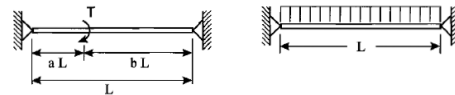
$$f_f = M_f / (S_y/2) = 13 / 180 * 2000 = 144.4 \text{ MPa}$$

Warping Stress Correction Factor (β) - Limited States Design In Structural Steel, Kulak & Gilmor (8th Ed)

-Assume worst case observation is on the mid of the span, therefore

$$\beta = 0.22$$

Table 5.2 – Correction factor β for the modified flexure analogy



Total Maximum Stress

$$\begin{aligned} f_T &= f_x + f_y + f_f * \beta \\ &= 87.3 + 102.2 + 144.4 * 0.22 \\ &= 221.3 \text{ MPa} \approx 0.63 F_y \end{aligned}$$

$$< \phi F_y = 0.9 F_y$$

μL	Concentrated torque					Uniform Torque (β at midspan)
	a = 0.1	a = 0.2	a = 0.3	a = 0.4	a = 0.5	
0.5	0.99	0.99	0.98	0.98	0.98	0.97
1.0	0.97	0.95	0.94	0.93	0.92	0.91
2.0	0.91	0.84	0.80	0.77	0.76	0.70
3.0	0.83	0.72	0.65	0.62	0.60	0.51
4.0	0.76	0.62	0.54	0.50	0.48	0.37
5.0	0.70	0.54	0.45	0.41	0.39	0.27
6.0	0.65	0.47	0.39	0.34	0.33	0.20
8.0	0.55	0.37	0.30	0.26	0.25	0.12
10.0	0.48	0.31	0.24	0.21	0.20	0.08

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- Of note, the result from this flexural analogy check is nearly identical to the Total Normal Stress from the JDB 2.F.06 Torsion Calculator. Therefore, we will conduct the remaining checks for the other section sizes with the calculator and assume the unity value should be below 0.9 for the section capacity.

8.5.1 Torison Calculator Input

- Reference HSC for the section dimensions and properties

Using methodology for RISA loadings and design bending moments as the hand check above.

Section	Self Wt (kN/m)	W' _{Pv} (kN/m)	W' _{Ph} (kN/m)	Span Length (m)	M _x (kN-m)	M _y (kN-m)	d (m)	e (m)	Torsion (kNm/m)
W10x22	0.321	6.72	1.68	4.572	17.6	4.4	0.258	0.18	0.30
W10x22	0.321	6.72	1.68	6.096	31.2	7.8			
W10x33	0.481	6.91	1.73	4.572	18.1	4.5	0.247	0.174	0.30
W10x33	0.481	6.91	1.73	6.096	32.1	8.0			
W10x49	0.715	7.20	1.80	4.572	18.8	4.7	0.253	0.177	0.32
W10x49	0.715	7.20	1.80	6.096	33.4	8.4			
W10x68	0.992	7.53	1.88	4.572	19.7	4.9	0.264	0.183	0.34
W10x68	0.992	7.53	1.88	6.096	35.0	8.7			
W12x26	0.38	6.79	1.70	7.62	49.3	12.3	0.31	0.206	0.35
W12x40	0.58	7.03	1.76	7.62	51.0	12.8	0.303	0.202	0.36
W12x40	0.58	7.03	1.76	9.144	73.5	18.4			
W12x53	0.774	7.27	1.82	7.62	52.8	13.2	0.306	0.204	0.37
W12x53	0.774	7.27	1.82	9.144	76.0	19.0			
W12x65	0.95	7.48	1.87	7.62	54.3	13.6	0.308	0.205	0.38
W12x65	0.95	7.48	1.87	9.144	78.2	19.6			
W14x43	0.627	7.09	1.77	7.62	51.5	12.9	0.347	0.224	0.40
W14x43	0.627	7.09	1.77	9.144	74.1	18.5			

8.6 Purlin Selection Summary (In-House)

Section (AISC)	Columns @ 15'				Columns @ 20'				Section (CISC)
	UC _{Mx,y}	UC _V	Δ/L Ratio	UC _σ	UC _{Mx,y}	UC _V	Δ/L Ratio	UC _σ	
W10x22	0.279	0.048	1163	0.516	0.525	0.064	491	0.837	W250x33
W10x33	0.157	0.043	1632	0.273	0.279	0.057	688	0.449	W250x49
W10x49	0.096	0.037	2481	0.158	0.170	0.050	1046	0.261	W250x73
W10x68	0.071	0.027	3416	0.109	0.127	0.036	1441	0.182	W250x101
Section (AISC)	Columns @ 25'				Columns @ 30'				Section (CISC)
	UC _{Mx,y}	UC _V	Δ/L Ratio	UC _σ	UC _{Mx,y}	UC _V	Δ/L Ratio	UC _σ	
W12x26	0.625	0.071	429	0.959	x	x	x	x	W310x39
W12x40	0.341	0.059	620	0.523	0.512	0.070	359	0.719	W310x60
W12x53	0.227	0.051	828	0.351	0.332	0.061	479	0.484	W310x79
W12x65	0.218	0.047	1005	0.268	0.314	0.056	581	0.370	W310x97
W14x43	0.307	0.050	858	0.490	0.462	0.060	496	0.670	W360x64

- UC_{Mx,y}, UC_V & Δ/L Ratio from RISA model & UC_σ from MathCad Torsion Calculator

x - Section sizes are not recommended for those spans.